

Predicting Teacher Demand in Philippine Public Schools Using Machine Learning: A Comparative Study of Multiple Linear Regression, Decision Tree, and Random Forest Models

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Abstract— The growing demand for teachers in Philippine public schools highlights the need for data-driven approaches to educational planning and resource allocation. This study applies and compares three machine learning models: Multiple Linear Regression, Decision Tree Regression, and Random Forest Regression to predict teacher demand across 17 regions in the Philippines using data from School Year 2016–2017 to SY 2020–2021. The dataset integrates regional enrollment figures, number of schools, and number of deployed teachers from publicly available government educational statistics. Feature engineering techniques including log transformation, student-to-school ratio, and temporal variables were applied to improve model learning. The BARMM was excluded from training and set aside to test generalization to a region with a different educational profile. Results showed that Multiple Linear Regression achieved the best overall performance with an R^2 of 0.9987, an RMSE of 373.23, an MAE of 229.30, and a mean cross-validation R^2 of 0.9957, indicating that teacher demand across most regions follows a largely linear pattern. However, when applied to BARMM, Random Forest produced the closest predictions with an average error of 43.5%, outperforming the Decision Tree at 55.8% and MLR at 84.2%, demonstrating stronger generalization to regions with unique characteristics. None of the models showed signs of overfitting. These findings suggest that MLR is most effective for regions with known patterns, while Random Forest is better suited for regions with distinct profiles, supporting more proactive and equitable teacher demand forecasting in the Philippines.

Keywords—*teacher demand forecasting, multiple linear regression, decision tree regression, random forest regression, Philippine public education*

I. INTRODUCTION

Education plays a vital role in national development by promoting economic growth, social progress, and human capital formation. In the Philippines, the public education sector continues to face challenges related to teacher shortages, overcrowded classrooms, and unequal distribution of educational resources. According to the United Nations Educational, Scientific and Cultural

Organization and the International Task Force on Teachers for Education 2030, the global education sector faces a significant shortage of teachers, highlighting the urgent need for effective workforce planning and sustainable educational policies [1]. This issue is also evident in the Philippine public school system, where increasing student enrollment places greater pressure on schools and educators.

The shortage of teachers and increasing workload among public school personnel have become major concerns in the country. The Senate of the Philippines emphasized the importance of protecting teaching and non-teaching personnel from workforce reduction policies due to the growing demand for educational staff [2]. In addition, studies by the World Bank revealed that limitations in educational resources, infrastructure, and service delivery continue to affect the quality of public education in the Philippines [3]. Despite the classroom organization guidelines established by the Department of Education, many schools still experience overcrowded classrooms and high student-to-teacher ratios [4].

Research also shows that excessive workload and administrative responsibilities negatively affect the performance and well-being of Filipino public school teachers. According to H. O. Tarraya, workload policies and non-teaching duties contribute to stress and reduced instructional effectiveness among educators [5]. These challenges highlight the need for data-driven approaches that can support better educational planning and teacher allocation.

To address these issues, this study proposes the use of machine learning models to predict teacher demand in Philippine public schools using historical educational datasets. Variables such as student enrollment, number of schools, and regional educational statistics are analyzed to identify patterns and forecast teacher requirements. By applying predictive models including Multiple Linear Regression, Decision Tree Regression and Random Forest, the study aims to support more efficient educational planning and resource allocation.

Furthermore, this study aligns with United Nations Sustainable Development Goal (SDG) 4: Quality Education, which promotes inclusive and equitable quality education for all. Through the integration of machine learning and educational data analysis, this research seeks to contribute to

more effective and sustainable educational planning in the Philippines.

II. METHODOLOGY

This section describes the sources of the datasets used in the study, the data preprocessing and cleaning procedures performed, the feature engineering techniques applied, and the methods used for training and evaluating the machine learning models.

A. Dataset Description

This study uses the four datasets covering Philippine public school data from School Year (SY) 2016–2017 to SY 2020–2021. The first two datasets contain the number of public schools per region per school year. The third dataset contains the number of teachers deployed per region per school year, which serves as the value the models are trying to predict. The fourth dataset contains the number of students enrolled per grade level from Kindergarten to Grade 6 in public schools, which were added together to get the total enrollment per region.

In addition, all four datasets cover the same 17 regions: NCR, CAR, Ilocos Region, Cagayan Valley, Central Luzon, CALABARZON, MIMAROPA, Bicol Region, Western Visayas, Central Visayas, Eastern Visayas, Zamboanga Peninsula, Northern Mindanao, Davao Region, SOCCSKSARGEN, Caraga, and BARMM.

Furthermore, the datasets used in this study were gathered from available government data sources, including the Philippine Statistics Authority Philippine Statistical Yearbook and the Philippine Open Data Portal. These platforms provide official educational statistics and regional public school records used for data analysis and machine learning modeling.

B. Data Cleaning

Each dataset used different names for the same regions that is why the name-matching list was created to make sure that all the datasets used have the same region labels. In addition, some numbers in the raw files were written with commas, such as "12,345." These commas were removed so the values could be read properly as numbers. Lastly, The three cleaned datasets which are the schools, teachers, and enrollment, were then combined into one table. Each row in the final table represents one region in one school year, giving a total of five school years across 17 regions.

C. BARMM set aside

During the initial exploration of the dataset, all 17 regions including BARMM are examined and processed together. The data for BARMM was complete and usable, covering all five school years with no missing values across enrollment, school count, and teacher count. At first glance, there was no immediate reason to treat it differently from the other regions.

However, upon closer inspection, the BARMM's numbers behaved differently from the rest. The relationship between its enrollment, number of schools, and number of teachers did not match the patterns seen in the other 16 regions. Because of this, keeping BARMM in the training data would confuse the models. To avoid this, BARMM was taken out of the training and testing process and saved on

the side. The models were built using only the 16 remaining regions.

D. Feature Engineering

To improve predictive performance, several derived variables were created from the original datasets. A `Students_per_School` feature was computed to measure enrollment pressure and school capacity within each region. In addition, log transformations are applied to enrollment and teacher variables to reduce skewness and improve model stability.

Temporal trend variables were also introduced through `Year_c` and `Year_squared` features. These variables allow the models to capture both linear and nonlinear educational growth patterns over time, helping improve long-term forecasting accuracy.

Categorical regional variables were converted into numerical representations using one-hot encoding through regional dummy variables. This enabled the machine learning models to recognize regional differences in teacher demand and enrollment behavior.

E. Data Split and Scaling

The full dataset was split into a training and testing with the 80:20 split, 80% for training and the remaining 20% is for the testing. This split was used for three models so the result could be compared fairly. But before the model training, numerical features were normalized using `StandardScaler` to ensure that variables with larger numerical ranges would not dominate the learning process.

F. Models and Evaluations

Three models were built and compared:

Multiple Linear Regression - It was used as the baseline model for identifying linear relationships between enrollment-related variables and teacher demand.

Decision Tree - It was implemented to capture nonlinear relationships and improve interpretability through rule-based predictions.

Random Forest - It was used to improve predictive accuracy and reduce overfitting by combining multiple Decision Trees into a single, more stable and reliable model.

The models were evaluated using R^2 , RMSE, and MAE, and were also tested using 5-fold cross-validation on the training set which gives more reliable prediction of how well the model performs on data it has not seen.

III. EXPERIMENTAL DESIGN

This section presents the experimental setup and analysis of how different the three model performs on the dataset with the same train-test split.

Multiple Linear Regression - The model works by finding linear relationship between the variables and the `log_teacher`. The log transformation are applied to both enrollment and teacher because their values and different across different regions, with this transformation the relationship become more stable and consistent which made it easier for the model to learn. In addition, region dummies are included in order for the model to account the difference of teachers regardless of the enrollment. Furthermore, the centered year and squared year variables were added to

capture how teacher demand changed over time without causing numerical problems during calculation.

The model was evaluated on both the training and test sets and further validated using 5-fold cross-validation. The results were stable across all folds, showing that the model was learning general patterns rather than memorizing the training data. Overall, MLR produced high R^2 values and low error on both sets, making it a strong and reliable model for predicting teacher demand.

```
===== BASIC MLR RESULTS =====
R2 Train      : 0.9982
R2 Test       : 0.9987
RMSE          : 373.23
MAE           : 229.30
CV R2 Scores  : [0.9913 0.9959 0.9985 0.9957 0.9971]
Mean CV R2   : 0.9957
Std CV R2    : 0.0024
```

Fig. 1. MLR Evaluation Results

Decision Tree - It was used to capture the nonlinear relationships and complex interactions within the educational dataset.

In addition, the hyperparameters were tuned using GridSearchCV, and the best model configuration found are maximum depth of 4, a minimum of 4 samples required to make a split, and a minimum of 2 samples at each leaf. The model achieved an R^2 of 0.9890 on the training set and 0.9754 on the test set, with a train-test gap of 0.0135, a RMSE of 1,635.52, and a MAE of 1,258.39. The mean cross-validation R^2 across 5 folds was 0.9577 with a standard deviation of 0.0102, indicating acceptable but less stable performance compared to the other two models. The model was evaluated using the same regression metrics applied to the baseline model.

```
Best Parameters Found:
max_depth: 4
max_features: 1.0
min_samples_leaf: 2
min_samples_split: 4

===== DECISION TREE RESULTS (Tuned) =====
R2 Train      : 0.9890
R2 Test       : 0.9754
Train-Test Gap: 0.0135 (lower is better)
RMSE          : 1,635.52
MAE           : 1,258.39
CV R2 Scores  : [0.9629 0.9711 0.9544 0.9405 0.9596]
Mean CV R2   : 0.9577
Std CV R2    : 0.0102
```

Fig. 2. Decision Tree Evaluation Results

Random Forest - It was used to improve predictive accuracy and reduce overfitting by combining multiple Decision Trees into a single, more stable model.

In addition, the hyperparameters were tuned using GridSearchCV, and the best model configuration are 200 trees with no depth restriction, a minimum of 2 samples required to make a split, and a minimum of 1 sample at each leaf. The model achieved an R^2 of 0.9969 on the training set and 0.9925 on the test set, with a train-test gap of only 0.0044, a RMSE of 902.17, and a MAE of 601.14. The mean cross-validation R^2 across 5 folds was 0.9764 with a standard deviation of 0.0062, indicating stable but slightly more variable performance compared to the other two models. The model was evaluated using the same regression metrics applied to the other two models.

```
Best Parameters Found:
max_depth: None
max_features: 1.0
min_samples_leaf: 1
min_samples_split: 2
n_estimators: 200

===== RANDOM FOREST RESULTS (Tuned) =====
R2 Train      : 0.9969
R2 Test       : 0.9925
Train-Test Gap: 0.0044 (lower is better)
RMSE          : 902.17
MAE           : 601.14
CV R2 Scores  : [0.9811 0.975 0.9751 0.9663 0.9846]
Mean CV R2   : 0.9764
Std CV R2    : 0.0062
```

Fig. 3. Random Forest Evaluation Results

IV. COMPARATIVE RESULTS

This section presents the comparison of the best performing machine models based on their R^2 , RMSE, MAE, and cross-validation scores.

A. Best Performance

The three models compared in this study are Multiple Linear Regression (MLR), Decision Tree Regression, and Random Forest Regression. Among the three, MLR achieved the best overall performance with an R^2 of 0.9982 on the training set and 0.9987 on the test set with an RMSE of 373.23, an MAE of 229.30, and a mean cross-validation of 0.9957 with a standard deviation of only 0.0024. Random Forest came in second with an R^2 of 0.9969 on training and 0.9925 on testing, with an RMSE of 902.17, an MAE of 601.14, and a mean CV of 0.9764 with a standard deviation of 0.0062. For the last place, Decision Tree performed the lowest among the three, with an R^2 of 0.9890 on training and 0.9754 on testing, with an an RMSE of 1,635.52, an MAE of 1,258.39, and a mean CV of 0.9577 with a standard deviation of 0.0102.

MODEL COMPARISON – MLR vs Decision Tree vs Random Forest			
Metric	Basic MLR	Decision Tree	Random Forest
R ² Train	0.9982	0.9890	0.9969
R ² Test	0.9987	0.9754	0.9925
Train-Test Gap	0.0005	0.0135	0.0044
RMSE	373.23	1635.52	902.17
MAE	229.30	1258.39	601.14
Mean CV R ²	0.9957	0.9577	0.9764
CV Std	0.0024	0.0102	0.0062
BARM Avg Error%	84.2%	55.8%	43.5%

Fig. 4. Best Performance Evaluation Result

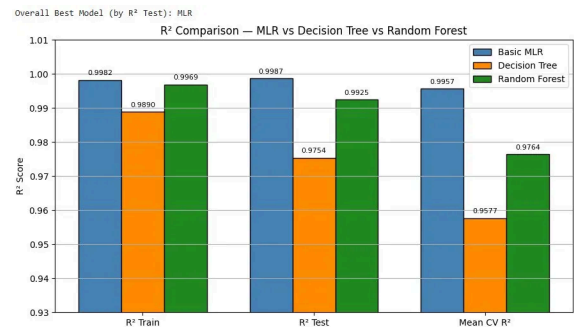


Fig. 5. R² Comparison Chart

Looking at the result, the MLR consistently produced the highest scores across all three evaluations. Random Forest placed second, showing strong but slightly lower performance, while the Decision Tree placed last on all three. The error comparison chart further confirms this ranking. MLR had the smallest prediction errors with an RMSE of 373 and MAE of 229, meaning its predicted teacher counts were closest to the actual values on average. The Decision Tree had the largest errors at 1,636 and 1,258, while Random Forest fell in between at 902 and 601.

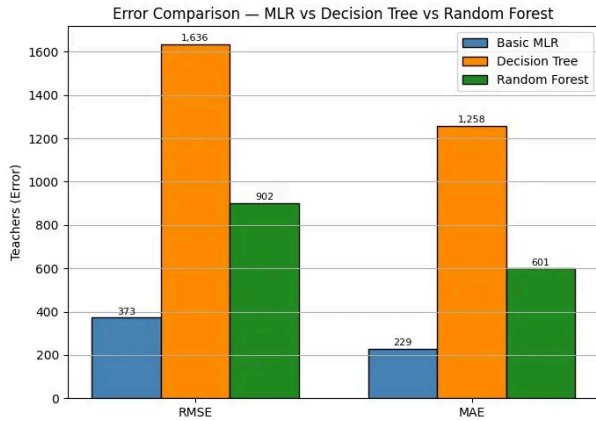


Fig. 6. Error Comparison Chart

The 5-fold cross-validation chart shows that MLR was not only the most accurate but also the most stable across all five folds, with scores between 0.9913 and 0.9985. Random Forest showed moderate version, with fold scores ranging from 0.9663 to 0.9846, while the Decision Tree was the least stable, dropping as low as 0.9405 in Fold 4 before recovering to 0.9596 in Fold 5. This pattern suggests that the Decision Tree is more sensitive to how the data is split, making it less reliable for consistent predictions.

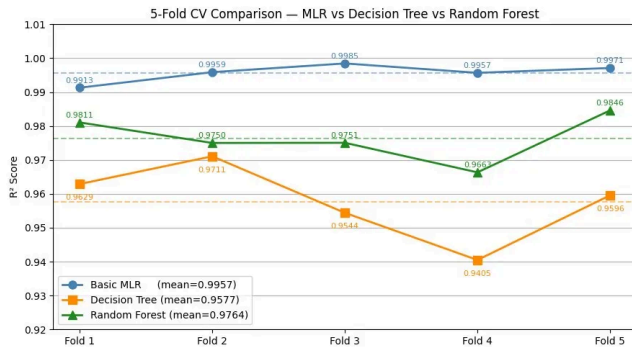


Fig. 7. 5 Fold CV Comparison Chart

B. BARMM Results

When the three models were applied to BARMM, which was completely excluded from training, all three overestimated the actual teacher counts across all five school years. This is expected given that BARMM's teacher deployment patterns and policies and government are different from the 16 regions the models learned from.

Among the three, Random Forest produced the closest predictions to the actual values, with an average error of 43.5%. The Decision Tree followed with an average error of 55.8%, while MLR had the largest BARMM error at 84.2%, significantly overestimating teacher counts in every school year.

This result reveals an interesting outcome, While MLR performed best on the 16 trained regions, it generalized the least well to BARMM. Random Forest, despite ranking second overall, showed the strongest ability to produce predictions that were closer to BARMM's actual numbers. This suggests that the nonlinear and ensemble-based nature of Random Forest made it better at adjusting to a region with a different educational profile, even without having seen it during training.

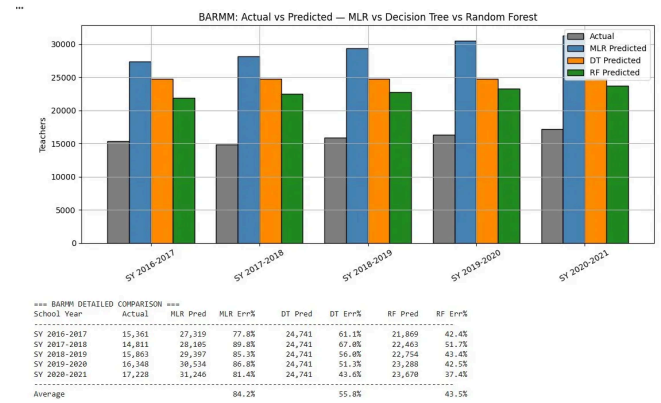


Fig. 8. BARMM Comparison Chart

V. CONCLUSION

This study applied three machine learning models: Multiple Linear Regression, Decision Tree, and Random Forest. Among the three models, Multiple Linear Regression performed the best on the trained regions, achieving the highest R^2 test score of 0.9987, the lowest RMSE of 373.23, the lowest MAE of 229.30, and the most stable cross-validation performance with a mean CV R^2 of 0.9957 and a standard deviation of only 0.0024. These results suggest that the relationship between enrollment-related variables and teacher demand follows a largely linear pattern across Philippine regions, making MLR the most suitable model for predicting teacher requirements in areas similar to those it was trained on.

Moreover, when the models were applied to BARMM, Random Forest produced the closest predictions with an average error of only 43.5%, outperforming both the Decision Tree at 55.8% and MLR at 84.2%. This indicates that while MLR excels at predicting teacher demand, Random Forest generalizes better to regions with different educational profiles.

Overall, this study demonstrates that machine learning models can effectively support teacher demand forecasting in the Philippine public school system.

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